

Prolog program for implementing the Towers of Hanoi

Aim

To write and execute a **Prolog program to implement the Towers of Hanoi** problem using recursion.

Algorithm

1. Start the program.
2. If $N = 1$, move the disk directly from the source peg to the target peg.
3. If $N > 1$, move $N-1$ disks from the source peg to the auxiliary peg.
4. Move the remaining largest disk from the source peg to the target peg.
5. Move the $N-1$ disks from the auxiliary peg to the target peg.
6. Print each disk movement using write/1 and nl/0.
7. Repeat the steps recursively until all disks are moved.
8. Stop.

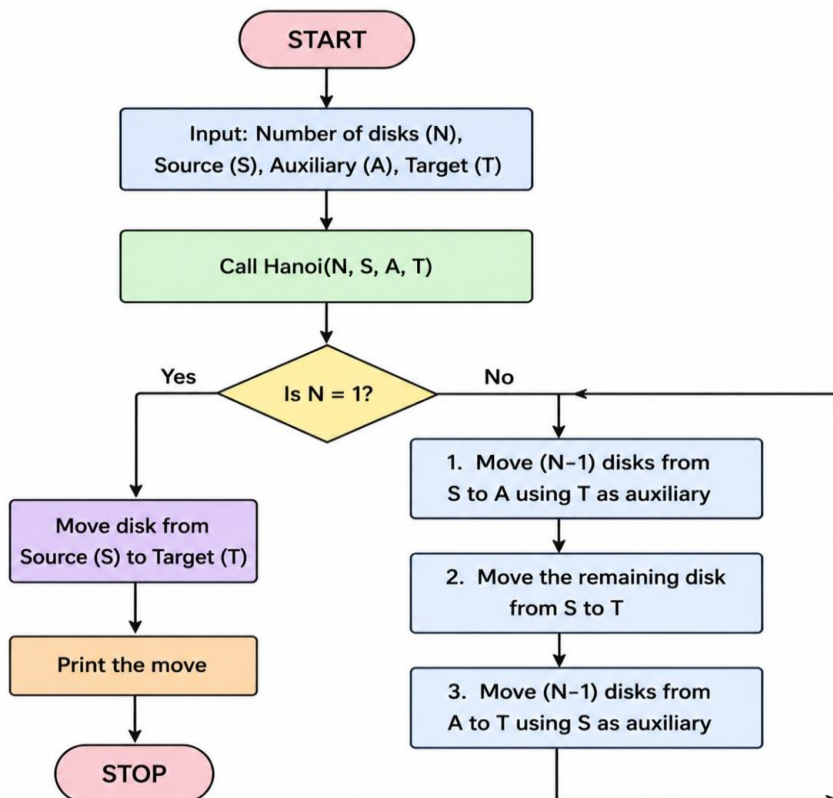
Program

% Program to implement Towers of Hanoi

```
hanoi(1, Source, Target, _) :-  
    write('Move the top disk from '),  
    write(Source),  
    write(' to '),  
    write(Target),  
    nl.  
  
hanoi(N, Source, Target, Auxiliary) :-  
    N > 1,  
    N1 is N - 1,  
    hanoi(N1, Source, Auxiliary, Target),  
    write('Move the top disk from '),
```

```
write(Source),  
write(' to '),  
write(Target),  
nl,  
hanoi(N1, Auxiliary, Target, Source).
```

Flow chart



Output

```
GNU-Prolog (AMC64, Multi-threaded version 9.2.9)  
File Edit Settings Run Debug Help  
Welcome to GNU-Prolog (threaded, 64 bits, version 9.2.9)  
GNU-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run '??- license' for legal details.  
For online help and background, visit https://www.gnu-prolog.org  
For built-in help, use '??- help(Topic)' or '??- apropos(Words)'.  
?-  
?- compile('C:/Users/Adrian/Desktop/Pr1/5.pl', compiled, 0.00 sec, 2 clauses).  
?-  
| hanoi(3, 'A', 'B', 'C').  
| Move the top disk from A to B  
| Move the top disk from A to C  
| Move the top disk from B to C  
| Move the top disk from B to A  
| Move the top disk from C to A  
| Move the top disk from C to B  
| Move the top disk from A to B  
| true.
```

Result

Thus, the **Prolog program for implementing the Towers of Hanoi was successfully executed**, and the required sequence of moves was obtained correctly.